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1. Introduction

Environmental issues and challenges have become a pressing global concern for a long time. Rachel Carson's book "Silent Spring" was published in 1962 and drew attention to the environmental impact of pesticides on both wildlife and humans. This, along with a series of environmental disasters, including the Santa Barbara oil spill and the love canal tragedy, mobilized people across the USA to participate in the environmental movement. People were increasingly concerned that their livelihoods were under threat from industrial pollution, nuclear power, toxic chemicals, habitat destruction and the depletion of natural resources. Environmental sociology emerged as a sub-discipline of sociology in the 1970s in response to growing awareness of environmental issues. It focuses on ecological problems in human societies as a result of industrialization, modernization, and capitalism. It also focuses on people's social responses and interactions to these problems. Given the complex interaction between society and the environment, scholars have adopted a specific analytical perspective from sociology that emphasizes the social conflicts and inequalities underlying environmental issues.

The primary scholarly focuses are based on three fundamental questions (Diekmann & Preisendörfer 2001, p. 19). The first question examines the mechanisms by which individual and institutional activities directly or indirectly impact the environment, and the motivations for adopting pro-environmental behaviors (PEBs). Previous studies have highlighted the importance of sociodemographic and psychological factors (Hunter, Hatch & Johnson 2004; Oreg & Katz-Gerro 2006; Secondi & Pirani 2011) in explaining why individual engage in PEBs. Those are related to e.g., gender, age, education level, environmental knowledge, and a series of pro-environmental attitudes, including risk perception, willingness to make sacrifice, etc.

The second question pertains to the dynamic changes in public awareness and attitudes towards environmental issues, particularly how people recognize and understand various environmental concerns as problems. By addressing these two questions, studies related to environmental attitudes and impact-oriented behaviors offer valuable insights into motivating individuals to participate in environmental protection efforts to address current problems and prevent further environmental degradation.

The third question aims to develop a deeper understanding of social and political responses to environmental challenges, such as the environmental movement and policy interventions to address environmentally damaging behaviors. In this context, the vital role of women in advancing sustainability cannot be underestimated. This is not only due to the growing

participation of women as environmentalists in the public sphere (Stein 2004, p. 2), with an increasing number of leaders being women (e.g., Sylvia Earle, Winona LaDuke, Jane Goodall, Greta Thunberg etc.), but also because environmental movements are more inclined to mobilize women, particularly mothers (Cousins 2021; Wilde & Parry 2022). An example of successful gender- and motherhood-based mobilization for environmental protection is the events in Love Canal, New York, in the 1970s. Local mothers played a pivotal role in organizing extensive and largely successful protests the toxic waste buried beneath their community. Lois Gibbs, a self-described “housewife who went to Washington”, took the lead after her children and those of her neighbors became ill because of the toxic environment (Newman 2001).

However, when examined through a feminist lens, the successful mobilization and positive environmental impact can reveal particular concerns. Wilde and Parry (2022) found that the zero-waste lifestyle movement, while promoting environmental awareness, also placed an increased domestic workload on women, further exacerbating the feminization of responsibility for sustainability. The movement's guidelines encourage participants to manage household waste independently, which requires the reorganization of domestic tasks such as cooking, cleaning, and shopping. Researchers have also observed that solutions to environmental issues increasingly rely on approaches such as green consumption and public transport use that are market- and individual-based, meaning that responsibility for environmental protection and risk management has shifted to individuals. Consequently, marginalized, and vulnerable groups, as well as women, are more likely to be disproportionately impacted and burdened by sustainable living practices (Bloodhart & Swim 2020; Cousins 2021; Lorenzen 2014; Vinz 2009; Wang 2016; Wilde & Parry 2022). For instance, promoting the use of walking, biking, or public transportation among women instead of personal cars might increase safety concerns for transgender, queers, as well as women and girls, more than for cisgender men and boys (Johnsson-Latham 2007). A large portion of PEBs, such as recycling, separating waste, and buying organic goods, are embedded in everyday routines, which have conventionally been associated with women's responsibilities (Vinz 2009; Wang 2016).

Despite this, the majority of research in environmental sociology has theorized women's support for environmental protection without exploring how women's disproportionate participation in environmental practices relates to broader structures such as patriarchy and political and economic systems (Banerjee & Bell 2007; Kennedy & Kmec 2018). As such, current discussions about promoting PEBs implicitly accept solutions that place the responsibility for addressing environmental problems on individuals, and women in particular. Furthermore, there is a lack of empirical research that examines how women's environmental engagement influences and intersects with their social and economic status (Dzialo 2017; Kennedy & Kmec

2018; Mackendrick 2014). In mainstream environmental sociology, gender is often considered alongside other socio-demographic factors (e.g., age, education, and income) and psychological variables (e.g., environmental attitudes) to predict engagement in PEBs. While a wealth of empirical research supports women's higher levels of environmental concern and behavior, particularly in the private sphere (Hadler & Haller 2011; Hunter, Hatch & Johnson 2004; Stern 2000), there is limited scholarship attempting to problematize and explain gender differences from a broader structural perspective (Davidson & Freudenburg 1996; Kennedy & Dzialo 2015; McCright & Xiao 2014). Therefore, gender differences in PEBs have the potential to be further explored and analyzed from different dimensions.

This study aims to provide empirical evidence on the role of gender in engagement in private and public PEBs and to understand how these gender differences are embedded in a broader social context. This will help to develop a more comprehensive and structural explanation for such differences. To achieve this, the study utilizes survey data from the International Social Survey Programme (ISSP) covering 13 countries and employs multilevel regression models that nest individuals (level 1) within countries (level 2). Multilevel models are advantageous as they enable the examination of country-specific characteristics and their influence on PEBs. These models consider both macro-level (country level) and micro-level (individual level) factors, as well as the effects between them through cross-level interactions (Ruppanner 2010). More specifically, in this study, the primary objectives are to systematically address the following inquiries in stages:

1. Does gender play a role in determining individuals' involvement in private and public PEBs within the countries under investigation?
2. To what extent can individual-level factors, such as environmental concern, and the broader contextual factor of gender inequality explain the patterns of engagement in private and public PEBs?
3. Does gender inequality play a significant role in shaping the feminization of responsibility for private or public PEBs?

By building upon existing research that explores individual-level and country-level variations in PEBs and incorporating a gender perspective, this study contributes to a quantitative analysis of how gender differences in PEBs are influenced by the context of each country.

The study is structured as follows: The subsequent section provides a systematic overview of gender and PEB. It presents a review of the literature that forms the basis of the analysis. Three widely accepted theoretical explanations for gender differences in PEB are introduced, along with an introduction to the application of multilevel analysis. Afterwards, the study addresses

the methods and data used to conduct the analysis. In this section, the issues related to the measurement and data analysis techniques are outlined. The findings of the statistical analysis are provided in the results section. The last section contains the discussion and limitations.

2. Gender and Pro-Environmental Behaviors

Pro-environmental behaviors have attained considerable attention in environmental research due to its potential to promote sustainable development. By examining variations in behaviors across diverse groups and contexts, researchers can gain valuable insights into how these behaviors are shaped by broader social inequalities. The purpose of this section is to provide a systematic review of existing research on PEBs, with a particular focus on gender issues in these behaviors. Before moving on to theoretical explanations of gender differences in PEBs, it is necessary to provide a clear definition of PEBs and how it has been discussed in the previous literature.

2.1. Definition of Pro-Environmental Behaviors

PEBs are distinguished by its focus on impact and action. It is important to differentiate PEBs from pro-environmental attitudes (PEA) and environmental concerns, which refer to “the degree to which people are aware of problems regarding the environment (e.g., climate crisis, biodiversity loss) and support efforts to solve them and/or indicate a willingness to contribute personally to their solution”(Dunlap & Jones 2002: 484). However, PEA as well as environmental concerns are intent-oriented, which implies that having environmental concerns or expressing a willingness to protect the environment does not necessarily translate into actual participation in PEBs.

There is no evident disagreement or debate on the definition of PEBs. Scholarships generally agreed that PEBs pertains to “behaviors that consciously seeks to minimize the negative impact of one’s actions on the natural and built world” (Kollmuss & Agyeman 2002). Examples of PEBs include recycling household waste, purchasing sustainable products such as local and organic goods, saving water and energy, and choosing alternative modes of mobility such as walking, cycling, or using electric vehicles. In addition, the definition has been broadened to include actions that not only reduce harm but also provide environmental benefits (Steg & Vlek 2009). This definition emphasizes the relevance of addressing issues like greenhouse gas (GHG) emissions and natural resource depletion in order to improve environmental conditions. From

a sustainability perspective, PEB includes actions that contribute to improving environmental sustainability (Mesmer-Magnus, Dechurch & Wax 2012). Therefore, PEBs take into consideration both self-interest and the well-being of others, future generations, different species, and entire ecosystems (Bamberg & Möser 2007). Furthermore, terms such as “responsible environmental behaviors”, “ecological behaviors”, “pro-ecological behaviors”, “environmentally friendly behaviors”, “environmentally responsible behaviors”, and “sustainable behaviors” have similar meanings to PEBs based on the definitions above.

PEBs are measured in several ways, but are generally discussed in the literature as either private- or public-sphere behaviors (Hunter, Hatch & Johnson 2004; McCright & Xiao 2014). Private PEBs typically include individual, everyday behaviors such as green consumption, energy and water conservation, and recycling. These behaviors, along with public PEBs that are community- or society-oriented such as participating in environmental movements, donating time or money, and signing petitions, are essential to promoting individual-level environmentalism.

2.1. Gender Differences in Pro-Environmental Behaviors and Explanations

Despite some contractionary (Arcury, Scollay & Johnson 1987), a substantial body of empirical evidence over several decades has shown that women generally have higher pro-environmental values and greater concern for the environment (Davidson & Freudenburg 1996; Dunlap & Mcstay 1983; Scannell & Gifford 2013; Zelezny, Chua & Aldrich 2000). When it comes to PEBs, however, research findings have been more inconsistent, particularly in the public sphere.

Zelezny et al. (2000) conducted a review of 13 studies conducted from 1988 to 1998, revealing that the majority of studies (9 out of 13) reported greater participation of women in environmental practices, 1 study indicated the opposite trend, and the remaining studies did not find significant evidence of gender differences. Hunter et al. (2004) examined gender differences in private and public PEBs in 22 countries and found that women were more involved than men in private PEBs in almost 70 percent of the countries studied. However, there was no significant evidence of gender differences in public PEBs. In addition, both women and men tend to be more involved in private environmental behaviors than in public ones. These observations are supported by recent studies conducted in various countries, including Australia (Organo, Head & Waitt 2013), China (Fan & Mendizabal Joffre 2020; Li, Wang & Saechang 2022; Xiao & Hong 2010), and the United States (Briscoe, Givens, Hazboun & Krannich 2019; McCright & Xiao 2014; Yates, Luo, Mobley & Shealy 2015).

The PEBs in the public sphere remain contentious. Some studies have found that men engage more in public PEBs than women (Davidson & Freudenburg 1996; Mohai 1992; Yates, Luo, Mobley & Shealy 2015), while others have found no differences (Briscoe, Givens, Hazboun & Krannich 2019; Hadler & Haller 2011; McCright & Xiao 2014). This study examines the gender disparities in the engagement of PEBs in both private and public spheres. It is expected that women will be more involved in private PEBs (Hypothesis 1) and that these gender differences will be moderated when it comes to engagement in public PEBs (Hypothesis 2).

Why are women more likely to participate in private PEBs, while gender differences disappear in public sphere? One widely accepted explanation is rooted in gender roles and socialization (Blocker & Eckberg 1997; Davidson & Freudenburg 1996; Eagly 1997; Hunter, Hatch & Johnson 2004; Trelohan 2022; Zelezny, Chua & Aldrich 2000). This explanation posits that individuals' attitudes and behaviors are shaped by the social and cultural norms and values they are exposed to throughout their lives. Specifically, the socialization process typically assigns individuals to gendered and socialized roles based on their physical sex. These roles and gender expectations directly and indirectly shape an individual's values and identity. For instance, boys are often trained to be more competitive, independent, and assertive, while girls are encouraged to prioritize caring for others, cooperation, altruism, and empathy (Eagly 1997; Gilligan 1993).

These different patterns of socialization lead to different expectations for future adults. Boys assume responsibility for their families, particularly in terms of material and social status attainment, whereas girls embrace caregiving roles. As a result, women tend to have higher levels of environmental awareness and concern, particularly with regard to health and safety issues. Their nurturing role and the general expectations associated with motherhood may contribute to their greater likelihood of adopting environmentally friendly practices and engaging in private PEBs (Hunter, Hatch & Johnson 2004; Torgler, Garcia-Valiñas & Macintyre 2008). In contrast, men are more likely to consider the economic benefits associated with engaging in PEBs, consistent with their societal role as "breadwinners" (Dünhoff, Stieß, Gigl & Birzle-Harder 2009).

The concept of gender maintenance identity provides an additional explanation for the differences in concerns and engagement between men and women. Men are often motivated to maintain their gender identity, which prompts them to avoid participating in PEBs associated with groups perceived as dissociative or feminine (Brough, Wilkie, Ma, Isaac & Gal 2016; McCreary 1994; Modlinska, Adamczyk, Maison & Pisula 2020). This effect is particularly apparent when it comes to consuming "green" products in public settings (White & Dahl 2006). For instance, the traditional association of meat consumption with masculinity has created

greater social and psychological barriers for males who choose to give up eating meat. On the other hand, females are more likely to find support from their friends and family when they become vegetarian or vegan (Modlinska, Adamczyk, Maison & Pisula 2020).

In addition, studies on identity-based motivation theory suggest that male participants prioritize personal success, competition, and the demonstration of skills, thereby valuing self-enhancement (Schwartz 2012). Female participants, on the other hand, place more emphasis on self-transcendence values and show more concern for social justice, uniting with nature, and environmental protection (Schwartz & Rubel 2005). As an example, women might be motivated by their responsibility to the environment and family members and feel guilty if they do not engage in PEBs (Trelohan 2022), while men adopt sustainable practices because they function as a costly signal associated with social status (Griskevicius, Tybur & Van den Bergh 2010). The more this behavior benefits his social status, the more likely he is to engage in it. Similarly, research on sustainable consumption by Costa Pinto et al. (2014) supported this theoretical consideration by showing that women engage in higher levels of sustainable consumption when their personal identity is salient, while men increase their levels of sustainable consumption to match women when their social identity is salient.

Explanations based on gendered socialization, social roles, and identity, combined with social norms and cultural influences, have inspired further empirical research on the factors that influence the adoption of PEBs and the factors that predict gender differences in engaging with PEBs. Rütgers (2021) investigated the association between compassion and PEBs. He found that women tend to have a greater preference for PEBs in food consumption and generally have higher compassion scores than men. In addition, women with higher levels of compassion tend to engage in more PEBs overall. Yates et al. (2015) utilized data from the 1994 and 2010 General Social Survey to explore the impact of sociodemographic variables, political ideology, and environmental concern on the relationship between gender and private and public environmentally responsible behaviors (ERBs). The findings indicated that women's higher levels of private ERBs were primarily driven by their more liberal political ideology and greater environmental concern compared to men. Moreover, it was found that full-time employment was significantly negatively associated with private ERBs for men, while it was significantly negatively associated with public ERBs for women. For this reason, women's higher engagement in private ERBs may be related to the association of these behaviors with traditional gender roles, particularly when women are not engaged in full-time employment. Furthermore, women's lower participation in public ERBs is not due to a lower propensity to translate their concerns into action. Instead, Yates et al. suggested that other broader constraints may hinder women's participation in public behaviors, and compared to men, women may be

less likely to engage in public ERBs as a means of enhancing their social status.

Whether measuring environmental concerns, levels of compassion, or employment status, previous research have predominately focused on socio-demographic factors and on values, beliefs, and other psychosocial constructs that are associated with PEBs reflecting variation at the individual level. However, as noted by Yates et al. (2015), there is a need to explore additional variables and take a broader perspective when explaining gender differences. This broader perspective involves considering the underlying context of gendered socialization, which is influenced by broader power structures such as patriarchy and political and economic frameworks. According to feminist literature, gendered socialization perpetuates and reinforces the current division of labor, with women often shouldering disproportionate responsibilities for reproductive tasks, unpaid household labor, and care work.

Despite the increasing dual-earner households, women continue to carry most of the housework, creating an additional “second shift” for working women (Hochschild 2003). Additionally, within neoliberal constructions of motherhood, mothers are assigned primary responsibility for the health and well-being of their children, including protecting them from environmental threats (Zivkovic, Warin, Davies & Moore 2010). As highlighted by Sandilands:

“Environmental privatization has a particular impact on women: it is women’s ‘traditional’ terrain that gets elevated as the apex of environmental behavior. On the surface, this may seem a positive step: a revalorization of women’s work, of ‘maternal’ behavior. But there is also a downside: if environmentalism is increasingly seen as household behavior, then it is women’s lives that come under the most intense scrutiny as the new private ecological morality comes into focus. And it is also a very particular conception of ‘women’ and ‘the family’ that gets invoked (Sandilands 1993, p. 46).”

Furthermore, the environmental movement has historically compelled women to fight for the well-being of their families in public. Existing societal expectations, on the other hand, often call for women to go home, emphasizing their participation in increased domestic work and consumption as a means of protecting the family, especially when a society's economic development stagnates. Thus, it is critical to recognize that this individualized approach to environmental protection, including preventative consumption, not only fails to provide sufficient overall protection, but also maintains existing gender inequalities and can exacerbate disparities among women of different racial and socioeconomic backgrounds. Gender differences in PEBs are thus tightly linked to underlying inequities in gender and power distribution.

To gain a broader understanding of gender differences, researchers have examined contextual factors in cross-national studies. Dzialo (2017) conducted a study using survey data from the

International Social Survey Programme (ISSP) in 2010, focusing on 30 industrialized countries. The study investigated the gender gap in private PEBs, including recycling, consumption of organic products, avoiding purchases for environmental reasons, frequency of avoiding car driving, and practices related to fuel, energy, and water conservation. The researcher took into account variables related to gender equality, national wealth, neoliberalism, and environmental concerns. Her findings revealed that in countries with higher gender equality in labor force participation, women tend to be more involved in PEBs compared to men. Drawing on theories from the sociology of work, she proposed that when there is greater parity between men's and women's labor force participation, traditional gender roles in the private sphere are reinforced. This occurs due to the ambiguity surrounding gender roles under the ideals of gender equality, leading to a reversion to more traditional gender roles (Kennedy & Kmec 2018).

While Dzialo acknowledged the impact of gender socialization on women's participation in private PEBs, she also questioned the “feminization of environmental responsibility” (Dzialo 2017), which refers to women's higher involvement in unpaid private PEBs compared to men. By incorporating structural and collective variables, there is an opportunity to examine and reevaluate how women are affected by their environmental protection efforts within the context of patriarchy and neoliberalism. However, Dzialo's study faced a significant limitation in terms of its analytical strategy. She measured the degree to which women reported higher or less engagement than men in six private-sphere PEBs compared to the overall participation in each country in order to quantify the feminization of environmental responsibility as a national-level variable. And she conducted ordinary least squares (OLS) regressions to investigate the impact of various national-level measures on feminization. Therefore, the study's sample size was equal to the number of analyzed countries, which was very small ($n = 30$). This limitation could result in issues such as limited statistical power, increased variability, violation of assumptions, overfitting, and limited generalizability. To obtain more reliable and robust regression results, Dzialo suggested including a broader and more representative sample from a more global perspective.

In sum, Dzialo has offered a unique perspective for future research that takes into consideration on the potential influence of both macro- and micro-level factors on gender differences in participation in PEBs. Based on this perspective, this paper examines how gender inequality at the country level may influence women's environmental behaviors. It is expected that in countries with greater gender inequality, women may take on more private environmental responsibilities which is often in line with traditional household responsibilities (Hypothesis 3). In terms of public environmental behavior, it is expected that in countries with greater gender inequality, women's participation may be less than that of men (Hypothesis 4).

2.3. Multilevel Analysis and its Application in Environmental Sociology

To overcome the limitations encountered in ordinary least squares (OLS) regression and improve the robustness of the findings, the study employs a multilevel regression model as the chosen analytical approach.

The multilevel regression model, also referred to by various names in statistical literature such as hierarchical linear model (HLM) (Lee & Bryk 1989), variance component model (Longford 1985), and random coefficient model (De Leeuw & Kreft 1986), is a statistical approach that allows for the analysis of data with a nested or hierarchical structure. In social research, it is common to study the relationships between individuals and the social contexts in which they are embedded. These social contexts can be seen as a system of individuals nested within groups, with individuals and groups defined at different levels of the hierarchy. The model recognizes that individuals within the same group are likely to be more similar to each other than individuals from different groups. Thus, the potential dependencies and variations that exist at both the individual level and the group level will be considered.

One classic work in developing and introducing the multilevel regression model is by Lee and Bryk (1989). They investigated the impact of school characteristics on student outcomes, specifically examining the relationship between individual-level factors (such as student characteristics) and school-level factors (such as school resources and teaching quality) in influencing student achievement. Their study revealed that both individual-level and school-level factors significantly influenced student achievement. Factors such as prior achievement and socioeconomic status at the individual level had an impact on student outcomes, while school-level factors such as resources and teaching quality also played a crucial role in explaining variations in student achievement across schools. This pioneering study contributed to advancing the theoretical and methodological framework of the multilevel regression model.

The study conducted by Guerin et al. in 2001 marked the first application of a multilevel regression model in the field of environmental sociology to examine cross-national variations in PEBs, with a specific focus on recycling participation (Guerin, Crete & Mercier 2001). The researchers utilized data from 15 European Union countries to explore the complex interplay between individual characteristics, societal and institutional factors, and national determinants in shaping recycling behavior. At the individual level, the study found that individuals with higher levels of global environmental concern and higher levels of participation in local environmental programs were more likely to recycle. In addition, higher levels of trust in

institutions were associated with a greater propensity to recycle. These findings contradicted the institutional trust hypothesis proposed by Davidson and Freudenburg (1996), which suggested that low levels of trust in institutions (in particular, those related to technology, science, and government) would lead to increased concern about environmental issues and may be associated with higher engagement in PEBs. Guerin et al.'s finding is also supported by Xiao and McCright (2014), who found that trust in science and government was associated with increased levels of environmental concern. Furthermore, the study identified two sociodemographic variables, namely education and income, as positively correlated with higher levels of recycling participation, suggesting that individuals with higher educational attainment and income levels were more likely to recycle.

At the country level, Guerin et al. (2001) concluded that the percentage of citizens involved in environmental organizations was a significant predictor of recycling participation. This variable alone accounted for approximately 45% of the observed variance between European Union countries in terms of public participation in recycling, which further highlighted the important role of collective variables at the national level for PEB analysis. Similarly, Pirani and Secondi (2011) revealed a positive association between a country's wealth and private PEBs. They further investigated the influence of national environmental investment, waste treatment percentage, and GDP on disposable items. However, the results regarding the impact of these variables on private PEBs were inconclusive, leading to ongoing debates in the field.

Beyond factors at the individual and country levels, exploration of cross-level interactions can offer additional information about e.g., gender inequality and gender differences (Fortin-Rittberger 2016; Fuwa 2004; Lagaert & Roose 2018; Pisano & Hidalgo 2014; Pisano & Lubell 2017; Yerkes, Roeters & Baxter 2020). Yerkes et al. (2020) examined how gender shapes the experiences and perceptions of leisure for men and women in different countries. They found that country-level factors such as traditional gender norms, inadequate childcare services, limited availability of paternity leave, and less political empowerment for women are directly related to lower quality leisure experiences for women compared to men. By examining the interaction between gender and country-level variables, they observed that in countries with greater gender equality, the gender gap in leisure quality decreases and in some cases reverses, highlighting the significant influence of social norms and institutional factors in shaping gender differences in leisure experiences and promoting gender equality. In the field of environmental sociology focusing on PEB, Pisano and Lubell (2017) conducted a cross-national study of environmental attitudes and behaviors and discovered a stronger positive effect of environmental attitudes on PEB in countries with a higher development index. Hitherto, the investigation of cross-level interactions is still at an early stage.

Building on previous research, the examination of cross-level interactions between gender and contextual or national variables allows for the investigation of the following key questions: in which specific contexts does the responsibility for environmental sustainability fall primarily on women? In particular, does higher gender inequality significantly affect women's participation in private and public PEBs, resulting in a potentially disproportionate burden on women to engage in PEBs? These questions prompt critical reflection on the implications and significance of gender inequality in women's participation in PEBs.

3. Data and Method

3.1. Data and Sample

The data from the 2020 Environment Module of the International Social Survey Program (ISSP) provides an opportunity to address the above questions and to empirically test the hypotheses developed in this study. The ISSP, as described by the official ISSP research group, is an annually rotating program that promotes cross-national collaboration and addresses topics of significant relevance to social science research. The environmental module focuses specifically on issues related to the environment, climate change, and environmental protection (ISSP Research Group 2022). The survey uses a questionnaire-based methodology, with face-to-face interviews, self-administered surveys, or mixed-mode techniques to collect data from highly representative national samples in 14 countries. The countries included Austria, Denmark, Finland, Germany, Hungary, Iceland, Japan, New Zealand, the Philippines, the Russian Federation, Slovenia, Switzerland, Taiwan, and Thailand. Nine of these countries are members of the Organization for Economic Cooperation and Development (OECD), suggesting that the observations in this study are concentrated in industrialized, high-income countries.

The age range of the individuals sampled is generally 18 and over, except in Finland where the sample includes individuals aged 15-74, and in Denmark where it includes those aged 18-79. The sample sizes for the countries varied, with Switzerland having the largest sample size of 4,280 individuals, and New Zealand having the smallest with 993 individuals. The majority of countries had sample sizes ranging from 1,000 to 2,000. Overall, the dataset comprises a total of 21,718 individuals from the 14 countries. It should be noted that the analysis excluded data from Taiwan ($n=1,822$) due to the unavailability of certain country-level data, such as the Gender Inequality Index. Additionally, the multilevel regression models only analyzed the observations with no missing value related to the variables used in the analysis. Therefore, the

final dataset utilized in the analysis consists of 13 countries and 17,761 individuals.

3.2. Dependent Variables: Private and Public Pro-Environmental Behaviors

To assess private PEBs, two survey items were employed to construct an index for this purpose: (1) How often do you make a special effort to sort glass or cans or plastic or papers and so on for recycling? (2) How often do you avoid buying certain products for environmental reasons? Participants responded on a scale from 1 (“always”) to 4 (“never”), and responses were recoded from 0 (“never”) to 3 (“always”). The index score was calculated as the average of the participants' ratings on these items. However, the scale reliability coefficient (cronbach’s alpha) for the index was found to be relatively low at 0.50. This may be attributed to the limited number of items used in constructing the index, as noted by previous studies (McCright & Xiao 2014; Yates, Luo, Mobley & Shealy 2015). To address this concern, a principle component analysis (PCA) was conducted to determine whether the two items were measuring the same underlying construct. The results of the PCA revealed that both items loaded on a single factor with an eigenvalue of 1.34, indicating a strong association between the two items. The factor loadings for both items were 0.82, further supporting their correlation. In cases where respondents reported that recycling facilities were not available in their community (2.6% of respondents), their responses to this particular item were excluded from the index calculation. For these respondents, therefore, the index score for private PEBs was determined only by their responses to the item measuring the frequency of avoiding certain products for environmental reasons.

Similarly, to measure public PEB, respondents' answers to four items were utilized: (1) Are you a member of any group whose main aim is to preserve or protect the environment? (2) In the last five years, have you given money to an environmental group? (3) In the last five years, have you signed a petition about an environmental issue? (4) In the last five years, have you taken part in a protest or demonstration about an environmental issue? The responses to these items were dichotomous, with “yes” coded as 1 and “no” as 0. The alpha reliability coefficient for the index was found to be 0.63. The PCA results indicated that all four items loaded onto a single factor with an eigenvalue of 1.92, and the factor loadings ranged from .60 to .75, indicating moderate to strong correlations with the principal component.

3.3. Independent Variables at the Individual-Level and Country-Level

The study used several variables to capture key sociodemographic characteristics and attitudes

of the participants. Gender was measured as a binary variable, with 0 representing male and 1 representing female. Sociodemographic variables included age (in years), years of education, and self-perceived social status (rated on a scale from 1 to 10, with 1 indicating very low status and 10 representing very high status).

Employment status was indicated by a dummy variable, with 0 indicating those without a job and 1 indicating those with a job. To further analyze the participants' employment status, a separate dummy variable was created to indicate whether they had a supervisory power at their workplace (0 = no supervisory position, 1 = supervisory position). In cases where individuals did not have a job, they were automatically assigned a value of 0.

Building on previous research, the study also considered the influence of willingness to make personal sacrifices and environmental concern on engagement in PEBs. Environmental concern was measured using the item “Generally speaking, how concerned are you about environmental issues?” on a scale ranging from 0 (not concerned) to 5 (very concerned). Willingness to make personal sacrifices was assessed through three items rated on a 5-point scale, ranging from 1 (very willing) to 5 (very unwilling). These items pertained to participants' willingness to pay higher prices and taxes, as well as their willingness to reduce their own living standards in order to protect the environment.

Gender Inequality Index (GII), developed by the United Nations Development Programme (UNDP) in 2021, was used as a measure to assess gender disparities across countries. Among the few available measures, the GII provides comprehensive insights into the inequalities and discrimination women face across multiple dimensions of life, including reproductive health, empowerment, and economic participation. The GII is constructed by combining several indicators, including maternal mortality rates, adolescent fertility rates, percentage of parliamentary seats held by women, educational attainment, and labor force participation. It is calculated on a scale of 0 to 1, with 0 representing gender equality and 1 representing extreme gender inequality across all measured dimensions.

3.4. Analytical Strategy

This study employs multilevel regression models (MRMs) to explore the influence of individual-level attributes on the participation in private and public PEBs across a variety of countries. The primary objective is to investigate whether potential differences in these effects can be attributed to gender inequality at the country level. In conceptual terms, the MRMs compute separate estimations of individual participation in private and public PEBs for each of

the 13 countries in the study. The resulting coefficients are subsequently treated as dependent variables in the country-level equations. This analytical approach facilitates the exploration of the random “country effect” linked to the average value of the dependent variable, providing invaluable insights into the cross-country variations observed.

Mathematically, the model at the individual level can be presented as

$$Y_{ij} = \beta_{0j} + \beta_{1j} * gender + \beta_{2j} * age + \beta_{3j} * education + \\ \beta_{4j} * status + \beta_{5j} * employment + \beta_{6j} * supervisor + \\ \beta_{7j} * environmental\ concern + \\ \beta_{8j} * willingness\ to\ make\ sacrifices + R_{ij}$$

where Y_{ij} denotes the dependent variable, capturing the engagement in PEB in either private or public spheres for individual i in country j , β_{0j} is the individual level intercept; β_{1j} to β_{8j} represent the fixed coefficients corresponding to the individual-level predictors, R_{ij} is the error term, assumed to be normally distributed with mean zero and variance σ^2 .

The country-level model can be formally presented as:

$$\beta_{0j} = \gamma_{00} + \gamma_{01} * GII + U_{0j}$$

In this equation, γ_{00} is the fixed effect or coefficient for the country-level intercept, γ_{01} is the effect of gender inequality on the β_{0j} which represents the country-level intercept capturing the baseline value of the model. U_{0j} is the random error term, assumed to be normally distributed with a mean of zero and a variance of σ^2 .

The MRMs outlined above regard the individual-level variables as fixed effects, inferring a consistent impact on the dependent variable across all countries. On the other hand, the gender inequality index is treated as a random effect, permitting the intercept to fluctuate across different GII levels. In a subsequent model, I examine the GII as both a fixed and a random effect to assess the average effect of gender inequalities on PEBs and to examine possible cross-country variations in this effect. Finally, I include cross-level interactions between the country characteristic and gender in the MRMs to determine how gender inequality might specifically affect women's participation in private and public PEBs, respectively.

Table 1. Descriptive Statistics of Individual-level Variables

	Men (N = 8,509)		Women (N = 9,253)		Population (N = 17,762)
Variable	Mean (M)	SD (M)	Mean (F)	SD (F)	p-value t-test gender difference
Independent variables					
Age	49.610	17.307	49.158	17.197	0.081
Environmental Concern	3.741	1.424	3.927	1.411	0.000
Willingness to Make Sacrifices	2.629	1.527	2.571	1.685	0.015
Employment (1 = yes)	0.657	0.475	0.599	0.490	0.000
Supervisory Position at Work (1 = yes)	0.336	0.472	0.169	0.375	0.000
Years of Education	13.241	4.076	13.100	4.033	0.020
Social Status	5.706	1.826	5.590	1.780	0.000
Dependent variables					
Participation in Private PEBs (1-4)	1.842	0.725	1.912	0.706	0.000
Participation in Public PEBs (0-1)	0.136	0.232	0.139	0.235	0.331

Source: ISSP 2020

4. Results

4.1. Descriptive Statistics

The descriptive statistics of the individual variables are presented in Table 1. The sample included in the analysis consisted of 9,253 female participants and 8,509 male participants. Gender differences were found to be statistically significant on several variables, including environmental concern, willingness to make personal sacrifices for environmental protection, employment status, supervisory power at work, years of education, self-perceived social status, and engagement in private PEBs. Overall, women showed significantly higher levels of environmental concern and engagement in private PEBs than men. However, a notable finding emerged regarding women's willingness to make personal sacrifices for environmental protection. Contrary to previous research indicating higher willingness among women (Stern, Dietz & Kalof 1993) or no significant gender difference (Yates, Luo, Mobley & Shealy 2015), the present study found that women were less willing to make personal sacrifices, such as paying higher prices and taxes or adjusting their current standard of living.

However, this unexpected finding does not contradict the fact that women are more actively involved in private environmental practices, such as recycling and avoiding environmentally

unfriendly products. Instead, it can be attributed to several factors. First, women may be more economically disadvantaged than men, given their lower employment rates, limited supervisory opportunities, fewer years of education, and lower self-perceived social status. Consequently, allocating additional financial resources to environmental causes may be more burdensome for women, which may further result in a lower willingness to make such sacrifices. Additionally, previous research has indicated that men derive a sense of pride and satisfaction from monetary contributions, which may make it a more rewarding choice for them in comparison to investing time or providing other support for environmental causes (Kemp, Kennett-Hensel & Kees 2013). The analysis also revealed that the gender difference in participation in public PEBs was not significant. This finding will be further explored and discussed in the subsequent section with multilevel regression results.

Table 2. Descriptive Statistics of Country-level Variables

Country	Gender Inequality Index (GII)	Mean Private PEBs	SD Private PEBs	Mean Public PEBs	SD Public PEBs	N
Austria	0.053	2.122	0.563	0.223	0.258	1223
Switzerland	0.018	2.206	0.471	0.232	0.282	3893
Germany	0.073	2.176	0.486	0.162	0.244	1551
Denmark	0.013	1.963	0.598	0.132	0.240	913
Finland	0.033	1.979	0.574	0.123	0.210	1077
Hungary	0.221	1.672	0.700	0.040	0.134	970
Iceland	0.043	1.994	0.533	0.202	0.267	845
Japan	0.083	2.004	0.587	0.035	0.105	1092
New Zealand	0.088	2.055	0.619	0.217	0.278	807
Philippines	0.419	1.547	0.683	0.049	0.131	1492
Russian Federation	0.203	1.023	0.907	0.055	0.141	1520
Slovenia	0.071	2.026	0.508	0.111	0.194	1067
Thailand	0.333	1.297	0.733	0.052	0.153	1312

Sources: Human Development Report 2021-22¹, ISSP 2020

Table 2 demonstrates the descriptive statistics of country-level variables. Among the 13 countries included in the study, Denmark had the lowest GII, with an impressive score of 0.013, indicating low gender inequality. On the other hand, the Philippines had the highest GII of 0.419, reflecting significant gender inequality across multiple dimensions. The average GII for all countries in the study was 0.127.

The average levels of private and public PEBs participation for all countries were 1.851 and

¹ See <https://hdr.undp.org/data-center/thematic-composite-indices/gender-inequality-index#/indicies/GII> [21.06.2023]

0.126, respectively. Among the countries, the Russian Federation had the lowest level of private PEBs involvement, with an average score of 1.023. On the other hand, Japan showed the lowest level of public PEBs participation, with a mean score of 0.035. People in Switzerland reported highest PEBs engagement in both private and public sphere.

Table 3. Direct and Cross-Level Interaction Effects on Engagement in Private Pro-environmental Behaviors

	Model 1		Model 2		Model 3	
Predictors	Estimates	SE	Estimates	SE	Estimates	SE
Intercept	0.912 ***	0.090	1.127 ***	0.097	1.126 ***	0.097
Individual level						
Gender (1 = female)	0.090 ***	0.009	0.090 ***	0.009	0.093 ***	0.012
Age	0.006 ***	0.000	0.006 ***	0.000	0.006 ***	0.000
Environmental Concern	0.066 ***	0.003	0.066 ***	0.003	0.066 ***	0.003
Willingness to Make Sacrifices	0.042 ***	0.003	0.042 ***	0.003	0.042 ***	0.003
Employment (1 = yes)	0.003	0.010	0.003	0.010	0.003	0.010
Supervisory Position At Work (1 = yes)	0.020	0.011	0.020	0.011	0.020	0.011
Years of Education	0.010 ***	0.001	0.010 ***	0.001	0.010 ***	0.001
Social Status	0.015 ***	0.003	0.015 ***	0.003	0.015 ***	0.003
Country level						
GII			-1.681 **	0.518	-1.669 **	0.519
Cross-level interaction						
Gender : GII					-0.023	0.069
ICC	0.217		0.134		0.134	
N(groups)	13 _{GII}		13 _{GII}		13 _{GII}	
N(observations)	17762		17762		17762	
Marginal R²	7.5%		21.2%		21.2%	
Conditional R²	27.6%		31.7%		31.7%	

Note: * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

SE: Standard Error

GII: Gender Inequality Index

Source: ISSP 2020

4.2. Men and Women's Pro-Environmental Behaviors across Countries

Table 3 presents the results of the multilevel regression models examining private PEBs. Consistent with previous research, the results confirm that women exhibit higher levels of engagement in private PEBs compared to men. This gender difference remains statistically significant after controlling for other individual-level factors, including age and environmental concern etc., supporting the first hypothesis. While employment status and supervisor status were not found to be significantly associated with private PEBs, the remaining variables provide a descriptive profile of individuals who are more likely to engage in private PEBs. Specifically,

they tend to be women, have higher levels of environmental concern, are older, have a greater willingness to make personal sacrifices for the environment, have more years of education, and perceive themselves as having a higher social status.

Model 1 elucidates that when solely considering the fixed effects of individual-level variables and the random effect of gender inequality index (GII), approximately 8% of the total variance in engagement in private PEBs can be attributed to the fixed effects alone. Overall, this model accounts for 27.6% of the total variance. In Model 2, when GII is introduced as both a fixed and random effect, the explanatory power increases to 31.7%. GII emerges as a significant factor of individuals' participation in private PEBs: in countries characterized by higher levels of gender inequality, individuals tend to exhibit lower engagement in private PEBs. However, the estimate for GII has a relatively high standard error of 0.52, indicating a considerable amount of uncertainty around this estimate. Consequently, the precision of this estimate is relatively low compared to other variables in the model. The study discussed the issue related to the value of the standard error in detail towards the end. Model 3 explores whether women in particular bear greater responsibility for practicing private PEBs within countries with higher levels of gender inequality (Hypothesis 3). However, the hypothesis is rejected, as no significant evidence of an interaction effect between gender and GII is observed.

Moving on to public PEBs as explored in Table 4, the findings reveal some intriguing facts. In Model 4, which controls for individual-level variables, there is no significant difference between men and women in terms of public PEBs participation. In fact, the p-value for the estimate for gender is only slightly above the threshold at 0.0051. The explanatory power of Model 4 is much lower than that of the model for private PEBs participation, at about 15%. When considering the effect of gender inequality on public PEBs in model 5, a negative correlation is found between GII and public PEBs, with a slight increase in explanatory power to about 17%. Surprisingly, the results change in an interesting way when the cross-level interaction effect of gender and country-level gender inequality on public PEBs is introduced. On the one hand, the significance of the gender effect becomes much stronger, indicating that women are also more involved in public PEBs, e.g., in environmental activities. On the other hand, the effect of gender inequality on public PEBs is not direct, as a significant interaction effect emerges. This suggests that in countries with higher levels of gender inequality, women tend to be less involved in public PEBs compared to men.

These findings support the hypotheses regarding public PEBs (Hypotheses 2 & 4) and provide valuable insights into the nuanced dynamics of gender, gender inequality, and public PEBs. While gender inequality plays a role in both private and public PEBs, the interaction effect

between gender and gender inequality appears to shape patterns of public engagement differently. Women's increased participation in public PEBs may indicate their active role in environmental advocacy, while the context of gender inequality influences the extent of women's involvement in public action.

Table 4. Direct and Cross-Level Interaction Effects on Engagement in Public Pro-environmental Behaviors

	Model 4		Model 5		Model 6	
<i>Predictors</i>	<i>Estimates</i>	<i>SE</i>	<i>Estimates</i>	<i>SE</i>	<i>Estimates</i>	<i>SE</i>
Intercept	-0.103 ***	0.020	-0.071 **	0.025	-0.076 **	0.025
Individual level						
Gender (1 = female)	0.006 ^a	0.003	0.006 ^a	0.003	0.016 ***	0.004
Age	-0.000 **	0.000	-0.000 **	0.000	-0.000 **	0.000
Environmental Concern	0.022 ***	0.001	0.022 ***	0.001	0.022 ***	0.001
Willingness to Make Sacrifices	0.019 ***	0.001	0.019 ***	0.001	0.019 ***	0.001
Employment (1 = yes)	0.003	0.004	0.003	0.004	0.003	0.004
Supervisory Position At Work (1 = yes)	0.003	0.004	0.003	0.004	0.004	0.004
Years of Education	0.006 ***	0.000	0.006 ***	0.000	0.006 ***	0.000
Social Status	0.005 ***	0.001	0.005 ***	0.001	0.005 ***	0.001
Country level						
GII			-0.247 *	0.124	-0.208	0.125
Cross-level interaction						
Gender : GII					-0.075 **	0.025
ICC	0.077		0.062		0.062	
N(group)	13 _{GII}		13 _{GII}		13 _{GII}	
N(observations)	17762		17762		17762	
Marginal R²	7.4%		11.3%		11.3%	
Conditional R²	14.5%		16.8%		16.8%	

Note: ^a $p=0.0051$ * $p<0.05$ ** $p<0.01$ *** $p<0.001$

SE: Standard Error

GII: Gender Inequality Index

Source: ISSP 2020

5. Discussion

The present study aimed to compare and explain gender differences in the adoption of PEBs across 13 countries, utilizing data from the 2020 ISSP Environment IV Module. By conducting a cross-national analysis, this study not only examined individual-level socioeconomic and psychological variations but also considered the broader structural inequalities related to gender across countries. The analysis focused on PEBs in the private sphere (recycling and avoiding products for environmental reasons) and the public sphere (involvement in environmental

groups, signing petitions for environmental protection, participating in environmental demonstrations, and donating money to environmental organizations). Using multilevel analysis, the current study examined whether there was a feminization of sustainable responsibility and, if so, whether the gender gap in private and public PEBs participation was influenced by gender inequality, measured by the gender inequality index provided by the UNDP in 2021. The overall findings of the analysis indicated a complex relationship between gender inequality and gender differences in PEBs, particularly in the public sphere.

Clear evidence of the feminization of private environmental responsibility was found across the observed countries, aligning with previous research that consistently shows women's higher participation in private-sphere PEBs compared to men on a cross-national scale (Dzialo 2017; Hunter, Hatch & Johnson 2004). Moreover, the study also revealed that gender inequality not only affects men but also women in relation to participation in private PEBs. Specifically, it was observed that both genders tend to engage less in private PEBs in countries with higher levels of gender inequality. Interestingly, when examining the interaction effect of gender and gender inequality, it is important to note that higher levels of gender inequality do not necessarily result in women shouldering a disproportionate burden of private sustainable responsibilities. On the one hand, this unexpected finding suggests that addressing gender inequality could be an effective strategy for promoting private PEBs among both men and women, highlighting the importance of addressing structural inequalities to promote environmental sustainability.

On the other hand, this finding raises an intriguing question that warrants further exploration: why don't women in countries with higher levels of gender inequality exhibit a disproportionate engagement in private sphere PEBs? One possible explanation for this observation is the economic disadvantages faced by women and their limited access to (environmental) education. The descriptive statistics of individual-level variables revealed that women tend to experience disadvantages across various aspects, such as education, economics, and willingness to pay or make personal sacrifices for the environment. In countries characterized by high levels of gender inequality, women may face resource constraints and limited opportunities for education, including environmental education. These constraints can impede their ability to effectively engage in PEBs, as they may lack awareness of environmental concerns, knowledge of potential solutions, and information on personal actions. Engaging in PEBs often requires a significant investment of time, knowledge, and financial resources. For example, finding alternative products or acquiring knowledge about environmental issues may involve extensive research and additional expenses. Thus, in societies where women may face more pronounced economic disadvantages, their capacity to participate in certain PEBs may be constrained due to a lack of

available resources and alternatives. Hence, women may face challenges in effectively participating, even if societal norms may assign private PEBs as “feminine” behaviors and encourage women to be involved in private environmental practices. Simultaneously, such societal norms may restrain men’s participation, leading to a lower environmental participation overall.

The results of the multilevel analysis for public PEBs support the hypothesis that women's higher level of environmental behavior is not limited to the private sphere, but also extends to the public sphere. It is noteworthy, however, that the gender difference in public PEBs only emerges when both individual- and country-level variables are considered in addition to the interaction effect. This finding offers a potential explanation for the contradictory results observed in earlier studies that focused exclusively on individual-level variables and relied primarily on ordinary least squares regression for their analyses. It underscores the importance of considering broader structural and contextual variations and applying diverse analytical approaches, thereby shedding light on the nuanced complexity of gender dynamics in PEBs.

Moreover, these findings present empirical evidence contributing to our understanding of the paradox where women, despite demonstrating higher environmental concern, risk perception, and engagement in private PEBs, appear reticent to translate these convictions into more visible action in the public sphere. While previous research has highlighted the important role of societal expectations, injunctive social norms, and gender socialization (Trelohan 2022; Zelezny, Chua & Aldrich 2000), the current study identifies structural gender inequality as a significant barrier to women's public engagement in environmental protection. Specifically, the findings suggest that women's propensity to translate their environmental passion into action is suppressed in countries with high levels of gender inequality.

This study has several limitations that need to be acknowledged. Firstly, the findings of this study may primarily be representative of wealthy industrialized nations, as most of the included countries are members of the OECD. While this subset of nations is highly relevant for examining private and public PEBs, it is important to recognize that generalizability to other countries and regions may be limited. To enhance statistical power and improve generalizability, future research should aim to include larger and more diverse samples of nations, encompassing a wider range of socioeconomic and cultural contexts.

Secondly, the ISSP module used in this study included only two questions related to private PEB and four questions related to public PEBs. This limited item pool hampers the researchers' ability to encompass the entire spectrum of PEBs, including various aspects and variations. As a consequence, the reliability scores obtained for the constructed indices of private and public

PEBs fell below the desired level. To address this limitation, future research should endeavor to incorporate a more comprehensive range of items that adequately reflect individuals' engagement in PEBs. For instance, in addition to recycling and consumption behaviors, it is essential to investigate other PEBs such as water and energy conservation, as well as mobility within the private sphere. By encompassing a broader set of behaviors, a more comprehensive understanding of the topic can be achieved.

Thirdly, in the Data and Sample section, it was mentioned that observations with missing data on the variables analyzed were excluded from the analysis. As a result, the original sample size of 19,870 was reduced to 17,762. Although the remaining sample size were still large, it may lead to several impacts, including a reduction in statistical power, which affects the precision and validity of the estimated regression coefficients. Moreover, if the missing data are not completely random, but rather missing at random or not missing at random, bias may be introduced. This may also affect the representativeness of the sample. Although minimal differences were observed when comparing the mean and standard deviation of variables before and after exclusion (as shown in Table 1, Table 2 as well as Table A1, Table A2 in the Appendix), this comparison alone does not guarantee the absence of potential bias resulting from the excluded observations. It is possible that bias still exists, especially if the missingness is related to unobserved characteristics. Therefore, it is recommended to employ more sophisticated imputation techniques in the future to ensure the robustness and validity of the results in multilevel regression analysis.

The higher standard error for the GII estimates observed in Model 2 and Model 3 (see Table 3), compared to the other estimates, indicates a wider confidence interval for the GII estimates and thus a higher degree of uncertainty around them. This could be attributed to the limited variability within the GII variables. It is noteworthy that 9 of the 14 countries examined have a particularly narrow GII range, between 0 and 0.1. The narrow range could be mainly due to the fact that the countries in the study are mostly well developed and have made significant efforts to improve gender equality. Future research could consider conducting separate analyses for countries with low and high GII values. This approach would allow for a more detailed examination of how gender inequality affects the variables under study in different contexts. In addition, it would be beneficial for future studies to include a more heterogeneous set of countries with a wider range of GII values. This would likely make the analyses more robust, leading to more reliable findings on the relationship between gender inequality and the variables of interest.

Despite the limitations, this study makes a valuable contribution to understanding patterns of

PEBs engagement among men and women, highlighting the influence of contextual factors and their interactions. Future research could delve deeper into examining the differential effects of factors within the concept of gender inequality to gain a more comprehensive understanding. For instance, different labor force participation rates and different working hours between men and women. Moreover, building upon these findings, subsequent studies can investigate the underlying mechanisms that drive PEBs, considering a broader range of contextual factors. The intricate interplay between micro- and macro-level influences emphasizes the importance of comprehensive approaches that address the fundamental challenges of gender inequality. By adopting such approaches, it is possible to foster increased participation and gender equality in PEBs, thus contributing to a more sustainable future.

7. References

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8. Appendix

Table A1. Descriptive Statistics of Individual-level Variables using Original Dataset (without removing observations with missing values)

	Men (N = 9,454)		Women (N = 10,416)		Population (N = 19,870)
Variable	Mean (M)	SD (M)	Mean (F)	SD (F)	p-value t-test gender difference
<i>Independent variables</i>					
Age	50.142	17.568	49.965	17.500	0.481
Environmental Concern	3.708	1.493	3.894	1.506	0.000
Willingness to Make Sacrifices	2.600	1.594	2.530	1.768	0.003
Employment (1 = yes)	0.650	0.477	0.591	0.492	0.000
Supervisory Position at Work (1 = yes)	0.339	0.473	0.171	0.377	0.000
Years of Education	13.197	4.163	13.050	4.084	0.014
Social Status	5.679	1.831	5.556	1.786	0.000
<i>Dependent variables</i>					
Participation in Private PEB (1-4)	1.847	0.723	1.922	0.704	0.000
Participation in Public PEB (0-1)	0.133	0.230	0.137	0.234	0.305

Note: Observations without gender information were excluded (n=26)

Source: ISSP 2020

Table A2. Descriptive Statistics of Country-level Variables using Original Dataset (without removing observations with missing values)

Country	Gender Inequality Index (GII)	Mean Private PEB	SD Private PEB	Mean Public PEB	SD Public PEB	N
Austria	0.053	2.120	0.568	0.223	0.259	1261
Switzerland	0.018	2.207	0.477	0.228	0.280	4263
Germany	0.073	2.178	0.490	0.162	0.244	1694
Denmark	0.013	1.967	0.599	0.124	0.235	1197
Finland	0.033	1.975	0.573	0.122	0.209	1137
Hungary	0.221	1.661	0.705	0.039	0.133	1001
Iceland	0.043	1.979	0.552	0.185	0.257	1150
Japan	0.083	2.031	0.590	0.038	0.119	1491
New Zealand	0.088	2.049	0.627	0.215	0.279	993
Philippines	0.419	1.548	0.683	0.049	0.132	1500
Russian Federation	0.203	1.022	0.908	0.054	0.140	1583
Slovenia	0.071	2.029	0.508	0.113	0.198	1102